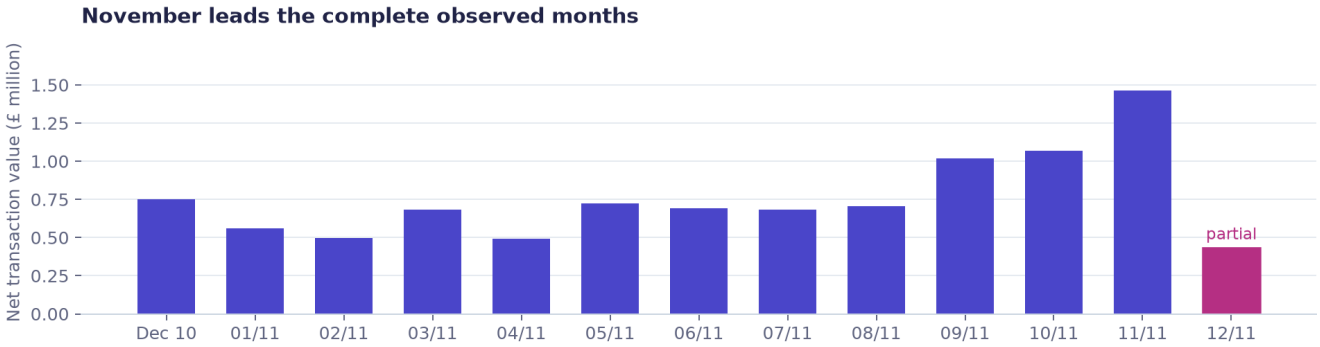
Retail sales & data quality audit

PERSONAL PORTFOLIO PROJECT | PYTHON / SQLITE / EXCEL
Actual public Kaggle / UCI transactions.

The task

Turn a transaction export into a useful sales report without dropping unknown customers, hiding cancellations or silently assuming repeated rows are errors.

Source lines	Valued net transactions	Excluded from valuation
541,909	£9,769,872.05	2,517 rows



Findings and practical implications

Valued net transaction value is £9,769,872.05: £10,666,684.54 gross positive sales less £896,812.49 in cancellations.

Missing customer IDs occur in 135,080 records. Identifying customers is not necessary to retain their £1,755,276.64 in gross sales.

Removing all exact repeated occurrences would reduce net value by £21,740.98 (0.2%). This is a sensitivity, not evidence that every repeat is an error.

Keep valid sales with unknown customers in aggregate reporting. Confirm line identifiers before deduplicating. Compare complete months and investigate large cancellations with the data owner.

Method, controls & limitations

TRACEABLE SOURCE ROWS | EXPLICIT RULES | RECONCILED TOTALS

Preserve and classify

Original rows remain in SQLite with source row numbers and exact-repeat fingerprints. Dates use month/day/year. Monetary line amounts use Decimal, rounded half up to pence. Eligible sales and C-invoice cancellations are retained; nonpositive prices and invalid records remain available for review.

Measure the effect of cleaning

Gross sales are £10,666,684.54 and cancellation value is £896,812.49. Excluding exact repeats changes net value from £9,769,872.05 to £9,748,131.07. Missing customer IDs affect customer analysis, not the validity of otherwise eligible sales.

Verification

22 automated test cases passed, including end-to-end repeat handling and a rebuild check. Seven full-data checks reconcile row counts, signed values, gross less cancellations, and monthly/country/product totals. Independent verification and workbook calculation checks are included in the project documentation.

Limits

December 2011 ends on December 9. These historical data describe one retailer. Special charge codes remain included in transaction values. Cancellation share is not a linked return rate. No cost or stock balance fields exist, so this project makes no profit, savings or inventory-availability claims.

SQL example: product ranking

```
WITH totals AS (
 SELECT sku, SUM(amount_pence) AS net_pence
 FROM valued_lines GROUP BY sku
)
SELECT DENSE_RANK() OVER (ORDER BY net_pence DESC) AS rank, *
FROM totals ORDER BY net_pence DESC, sku;
```

Source and attribution

Chen, D. (2015). *Online Retail*. UCI Machine Learning Repository. DOI: 10.24432/C5BW33. Downloaded from Kaggle: carrie1/ecommerce-data (October 15, 2026). UCI lists CC BY 4.0; the mirror lists Unknown. Full URLs, SHA-256 and transformation details are in SOURCES.md.